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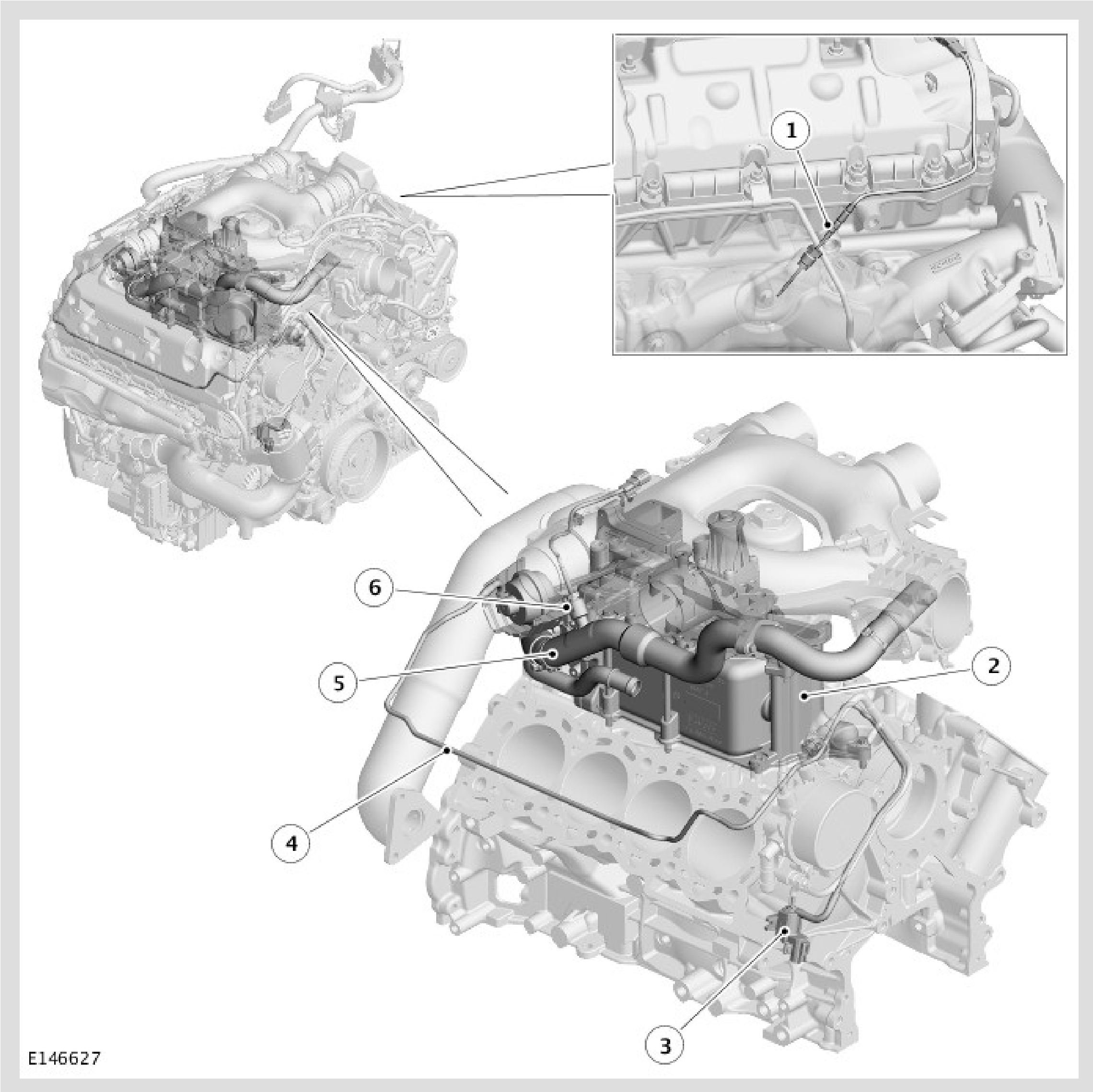
2015.0 RANGE ROVER (LG), 303-08

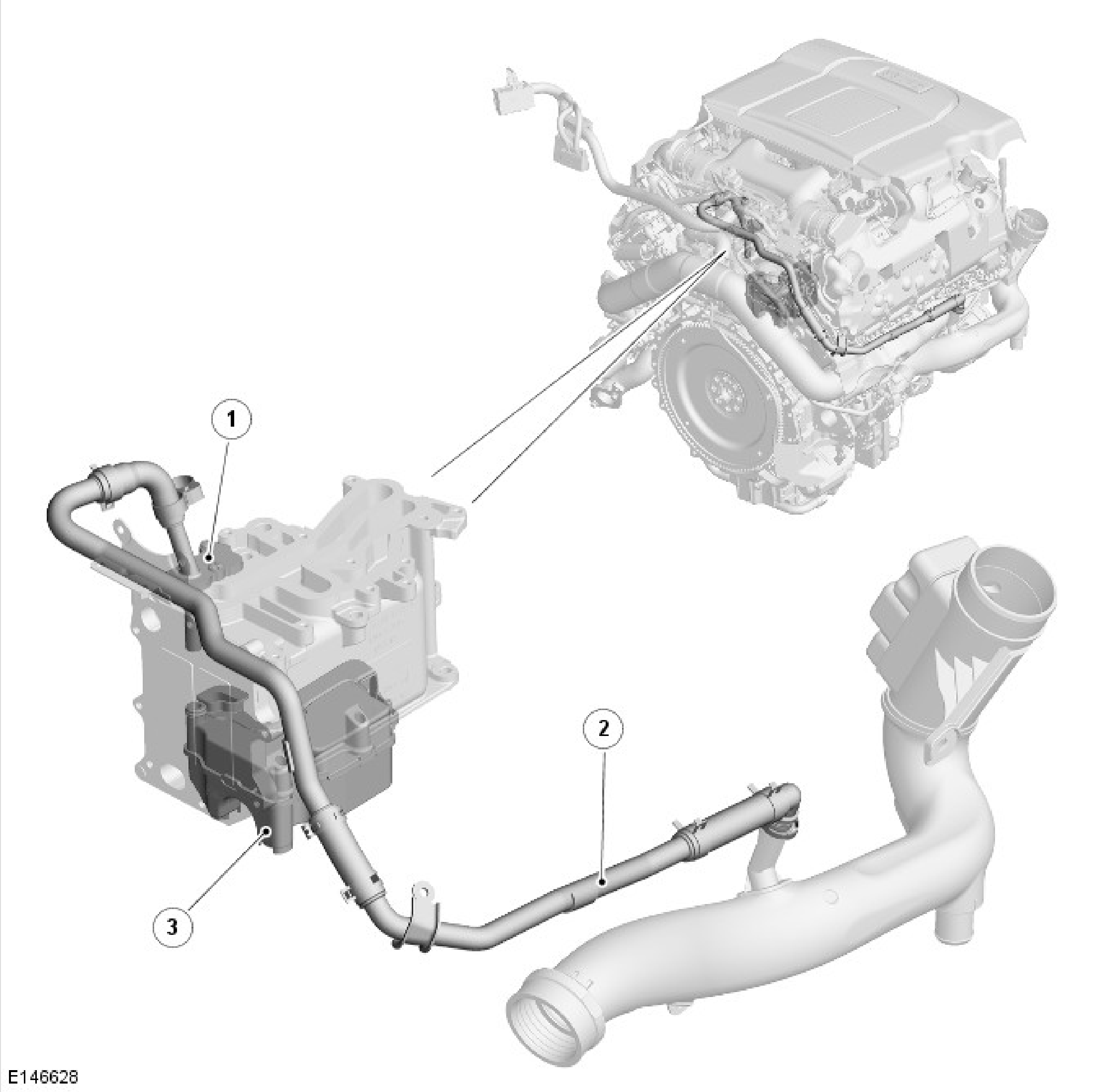
# ENGINE EMISSION CONTROL - TDV8 4.4L DIESEL

## COMPONENT LOCATION - EXHAUST GAS RECIRCULATION (EGR)




| ITEM | DESCRIPTION |
|---|---|
| 1 | Exhaust Gas Recirculation (EGR) inlet temperature sensor |
| 2 | Exhaust Gas Recirculation (EGR) cooler and valve housing |
| 3 | Exhaust Gas Recirculation (EGR) bypass valve solenoid |
| 4 | Vacuum tube |
| 5 | Exhaust Gas Recirculation (EGR) outlet tube |
| 6 | Exhaust Gas Recirculation (EGR) outlet temperature sensor |

## COMPONENT LOCATION - CRANKCASE VENTILATION




| ITEM | DESCRIPTION |
| --- | --- |
| 1 | Oil baffle and ventilation cap |
| 2 | Crankcase ventilation hose and tube assembly |
| 3 | Oil separator |

## OVERVIEW

### EXHAUST GAS RECIRCULATION (EGR) SYSTEM

The Exhaust Gas Recirculation (EGR) system is used to control the amount of exhaust gas being recirculated in order to reduce exhaust emissions and combustion noise. EGR is enabled when the engine is at normal operating temperature and under cruising conditions.

### CRANKCASE VENTILATION SYSTEM

The crankcase ventilation system ensures that all gasses emitted from the crankcase during engine running are separated from any oil particles and recirculated via the clean air induction system.

## DESCRIPTION

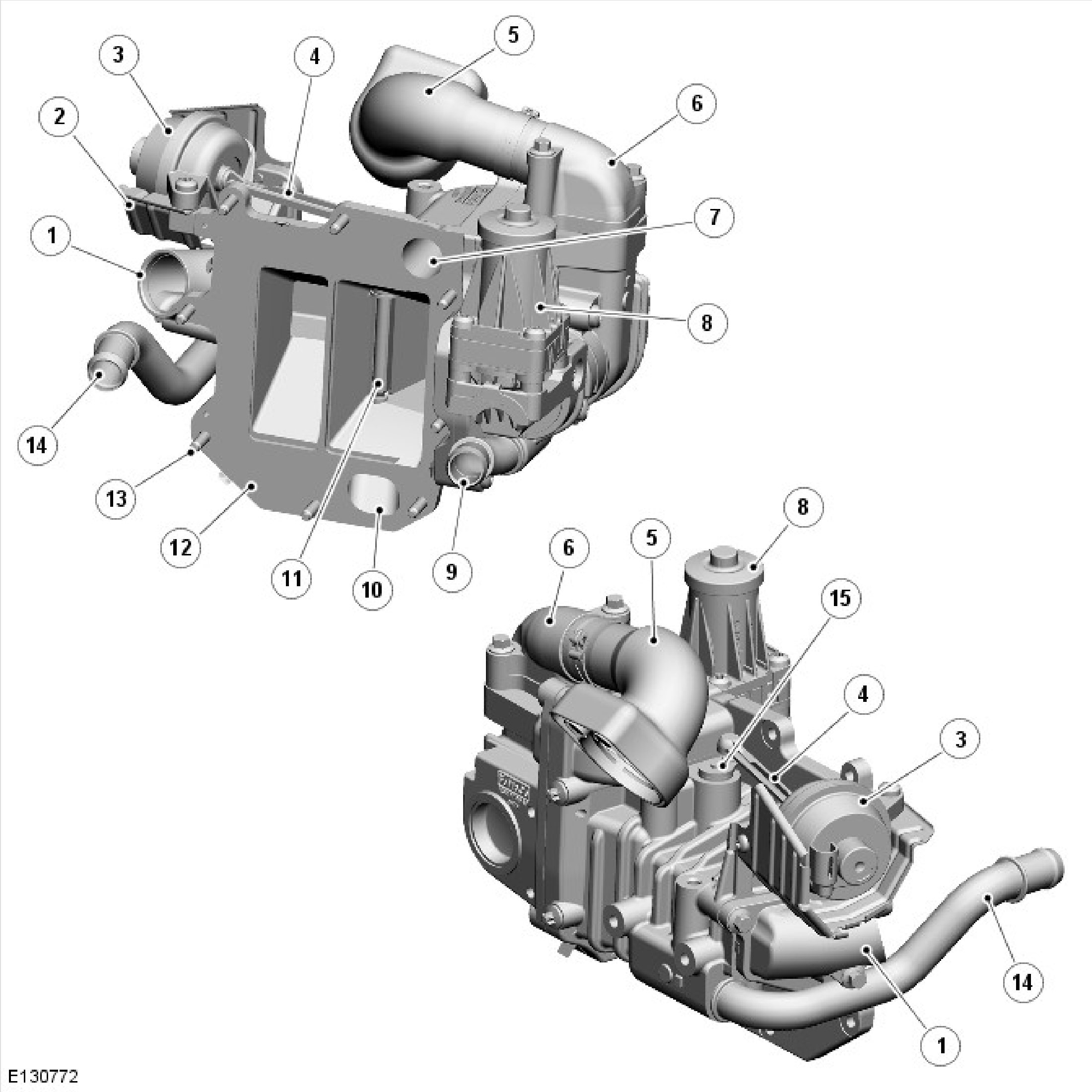
### EXHAUST GAS RECIRCULATION SYSTEM

The EGR system comprises an EGR cooler and valve housing assembly, which is attached to the cylinder block between the two cylinder banks. The EGR cooler and valve housing assembly incorporates an EGR valve and motor, an EGR cooler, and a cooler bypass valve and actuator.

The EGR valve housing is connected to the vehicle cooling system, via hoses. The EGR intake pipe is connected to the exhaust manifold crossover pipe at the rear of the engine. The exhaust gas passes through the cooler and is expelled through EGR outlet pipe into the intake manifold.

The EGR valve motor is a Direct Current (DC) stepper motor with integral position sensor, which is controlled by the Engine Control Module (ECM). The ECM uses the EGR valve motor to control the amount of exhaust gas being recirculated in order to reduce exhaust emissions and combustion noise. The EGR is enabled when the engine is at normal operating temperature and can operate at engine idle or under all part load operating conditions, in both the transient and steady states. EGR only operates in mono turbocharger mode and is shut-off before bi-turbocharger mode is initiated.

### EXHAUST GAS RECIRCULATION (EGR) VALVE HOUSING




| ITEM | DESCRIPTION |
|---|---|
| 1 | Exhaust Gas Recirculation (EGR) outlet pipe connection |
| 2 | Heat shield |
| 3 | Exhaust Gas Recirculation (EGR) vacuum actuator |
| 4 | Actuator rod |
| 5 | Exhaust Gas Recirculation (EGR) intake pipe |
| 6 | Exhaust Gas Recirculation (EGR) valve housing |

| | |
|---|---|
| 7 | Coolant gallery |
| 8 | Exhaust Gas Recirculation (EGR) valve motor |
| 9 | Exhaust Gas Recirculation (EGR) cooler engine coolant outlet |
| 10 | Coolant gallery |
| 11 | Bypass valve |
| 12 | Gasket |
| 13 | Attachment screws (8 off) |
| 14 | Exhaust Gas Recirculation (EGR) engine coolant intake |
| 15 | Bypass valve actuator arm |

The EGR valve housing is attached to the EGR cooler end tank housing. The valve housing is sealed to the end tank housing with a gasket and secured with 8 screws.

The EGR valve motor is secured to the valve body with 4 torx screws and a gasket. The valve motor has an electrical connector which connects into the engine wiring harness.

The EGR vacuum actuator is attached to the valve housing with 2 torx screws. A heat shield is positioned under the actuator to protect it from heat emitted from the EGR outlet pipe. The actuator is connected to the emissions vacuum system and is operated by a vacuum produced by the vacuum pump and controlled by an ECM controlled EGR bypass valve solenoid.

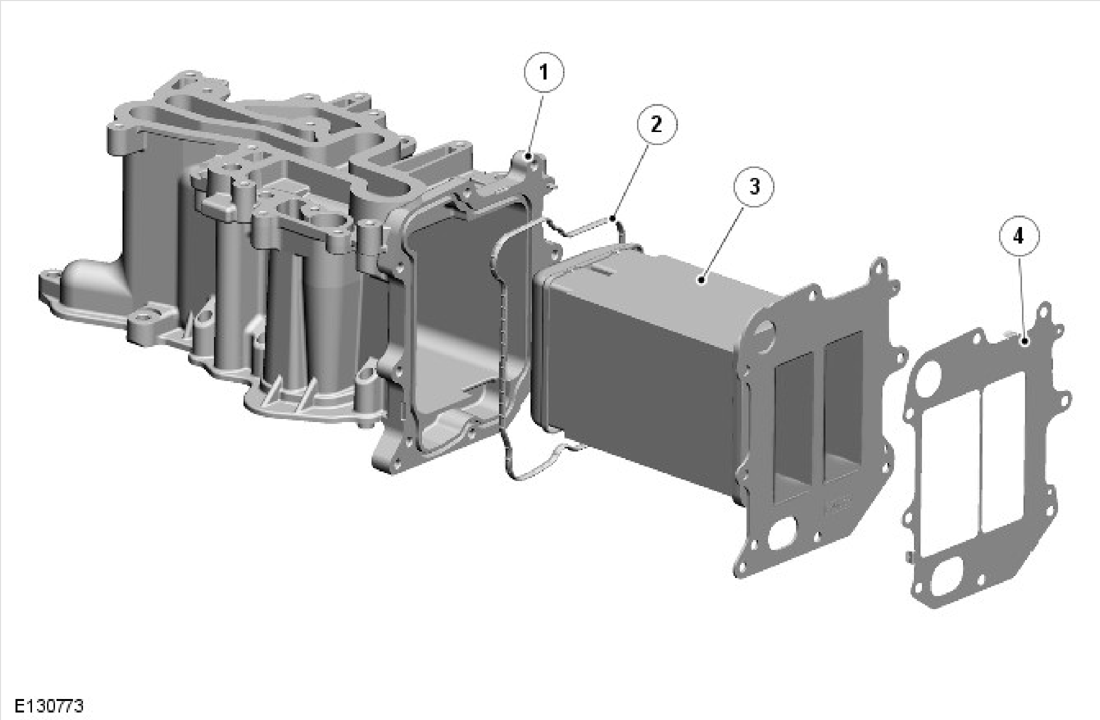
The EGR vacuum actuator is connected to the bypass valve in the valve housing by an actuator rod and arm. Linear movement of the vacuum actuator is passed through the rod to the arm, which converts the movement into rotary movement of the bypass valve. The bypass valve is either open or closed, so all exhaust gasses are either cooled by being directed through the cooler or bypass the cooler and pass directly into the EGR outlet pipe.

The EGR intake pipe is connected to the valve housing with two bolts and sealed with a gasket. The pipe between the valve housing and the crossover pipe connections is fitted with an insulating sock to reduce heat emissions from the pipe. The intake pipe is attached to the crossover pipe between the two exhaust manifolds and is secured with a clamp.

Exhaust gasses exit the valve housing via an EGR outlet pipe which connects into the intake manifold.

Two pipes allow for the connection of hoses from the engine cooling system. The two pipes allow for the engine coolant to flow into the valve housing, around the EGR cooler and out through the valve housing and back into the engine cooling system.

## EXHAUST GAS RECIRCULATION (EGR) COOLER



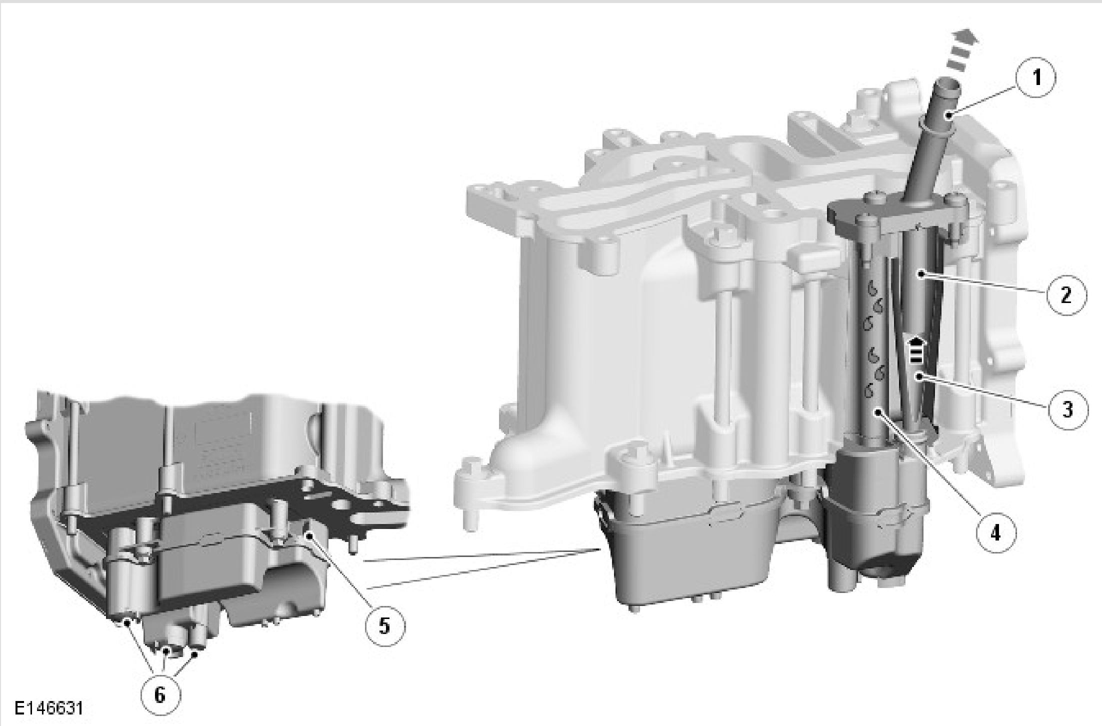

| ITEM | DESCRIPTION |
|---|---|
| 1 | End tank housing |
| 2 | Seal |
| 3 | End tank cooler |
| 4 | Gasket |

The EGR cooler assembly is located on the top of the engine and secured to the crankcase with 9 bolts. A gasket between the end tank housing and the crankcase prevents oil leakage. Three threaded bosses on the base of the end tank housing provide for the attachment of the crankcase ventilation oil separator. The hole which passes through the end tank housing allows for the crankcase ventilation tube location and connection with the oil separator. The top face of the end tank housing provides the location for the engine oil filter and cooler assembly.

The end tank housing has 8 threaded holes which allow for the attachment of the EGR valve housing. A gasket in the end tank housing seals the joint between the housing and the end tank cooler. A gasket seals the joint between the end tank cooler and the valve housing.

Two holes in the end tank cooler allow engine coolant from the thermostat housing to pass from the valve housing, circulate around the void between the end tank housing and the end tank cooler, then return to the valve housing before being passed back to the thermostat housing.

## CRANKCASE VENTILATION




| ITEM | DESCRIPTION |
| --- | --- |
| 1 | Ventilation cap |
| 2 | Oil baffle |
| 3 | Gas outlet channel |
| 4 | Oil drain channel |

| | |
|---|---|
| 5 | Gas intake |
| 6 | Oil drain valves |

The crankcase ventilation system comprises an oil separator and a ventilation tube and hose assembly.

The oil separator is attached to the base of the EGR cooler, between the two banks of the cylinder block. An opening in the side of the oil separator provides an intake for gasses from the crankcase. Drain valves in the base of the oil separator provide outlets for the oil. Two spigots on top of the oil separator locate in a gas outlet channel and an oil drain channel in the base of the EGR cooler. The other end of the channels are fitted with an oil baffle and ventilation cap. The ventilation tube and hose assembly is connected between the ventilation cap and the intake duct of the primary turbocharger.

## OPERATION

### EXHAUST GAS RECIRCULATION SYSTEM

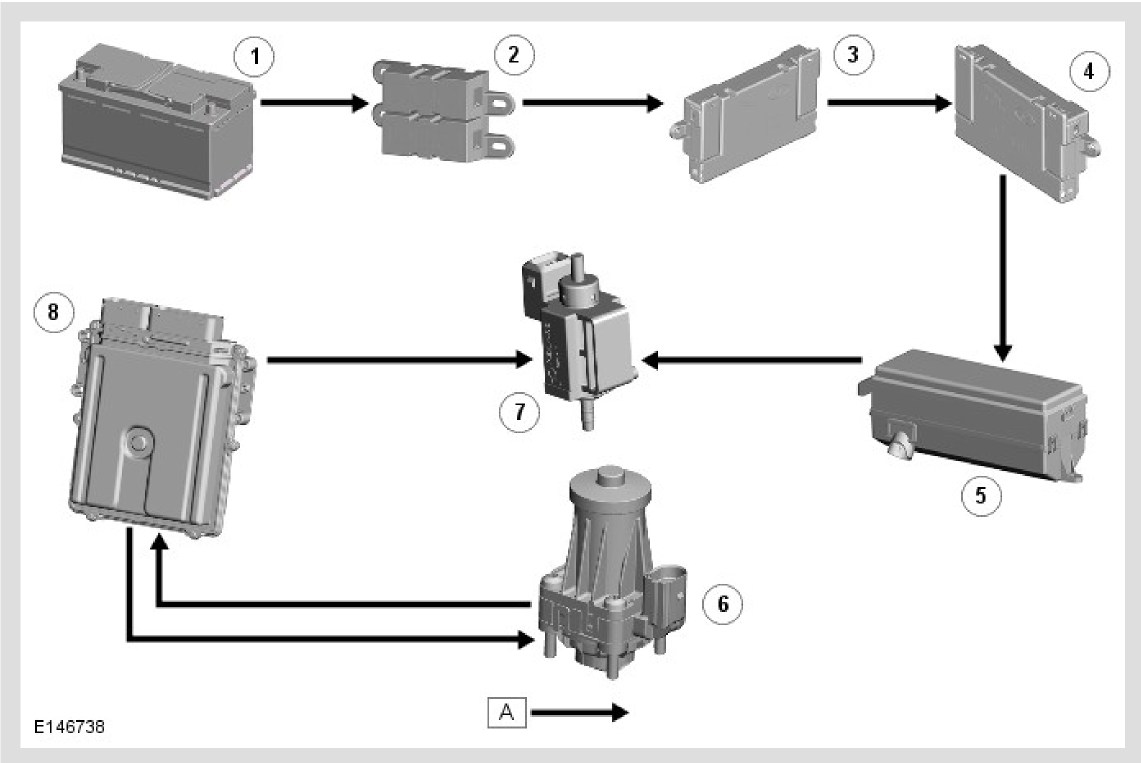
If small volumes of fuel are injected into a combustion chamber full of pure air, the effect is to create a very lean mixture. This burns at a high temperature, which in turn causes the excess oxygen in the mixture to combine with the naturally occurring nitrogen in the air to create Nitrogen Oxide (NOx), a noxious class of pollutants associated with acid rain. This is a particular problem for diesel engines at low to medium loads. Exhaust gas is blended into the intake air charge to create the cylinder charge. As the exhaust gas effectively contains no oxygen, it prevents the formation of a very lean mixture, so lowering combustion temperatures and minimizing the formation of NOx.

At low engine speeds and loads, over 50 percent of the cylinder charge can be made up of recycled exhaust gas. This is routed from the exhaust manifolds through the EGR coolers before being supplied to the electric throttle. The volume of exhaust gas added to the intake charge is regulated by the EGR valves, actuated according to precise engine speed and load maps by the ECM.

### CRANKCASE VENTILATION

Clean air being drawn into the primary turbocharger when the engine is running creates a vacuum in the ventilation tube and hose assembly. This vacuum in turn creates a vacuum in the oil separator and draws gasses from the crankcase into the separator. The gasses pass through the oil separator, where oil particles separate from the gasses. The gasses are then drawn through the gas outlet channel in the EGR cooler, where the oil baffle removes any remaining oil particles, which drain back into the oil separator. The gasses then pass through the ventilation tube and hose assembly to the clean air duct, and are mixed with the clean air being drawn into the turbocharger. The oil drains out of the oil separator through the drain valves, and down through the cylinder block into the oil pan.

## CONTROL DIAGRAM




## A = HARDWIRED.

| ITEM | DESCRIPTION |
|---|---|
| 1 | Battery |
| 2 | Battery Junction Box 2 (BJB2) |
| 3 | Battery Junction Box (BJB) |
| 4 | Auxiliary Junction Box (AJB) |
| 5 | Engine Junction Box (EJB) |
| 6 | Exhaust Gas Recirculation (EGR) bypass valve solenoid |
| 7 | Exhaust Gas Recirculation (EGR) valve motor |

| | |
|---|---|
| 8 | Engine Control Module (ECM) |